

# RegimeLens

AI Tool for Asset-Level Market Regime Detection

## Reflections:

Through the development of RegimeLens using Product Vision and User Story Mapping, I realized that Agile thinking is fundamentally about disciplined scope control and value prioritization rather than technical sophistication.

Initially, my inclination was to focus on advanced machine learning architectures such as transformers or reinforcement learning. However, once I structured the Product Vision Board and defined clear user personas, it became evident that interpretability and clarity deliver more value than predictive complexity. This shift in perspective demonstrates one of Agile's core principles: customer value over technical elegance.

Building the backbone features forced me to think in terms of user journey rather than system components. Instead of asking, "What models can I implement?", I began asking, "What does the user need to understand first?" This reorientation transformed the product from a predictive system into an explainable analytical tool.

The MVP slicing process was the most intellectually valuable step. By drawing a horizontal cut across the story map, I had to distinguish between value-enabling features and value-enhancing features. This exercise highlighted the importance of incremental delivery, risk reduction, and early validation, pillars of Agile delivery.

Overall, this assignment reinforced that Agile development is not about moving fast, it is about building the right thing, at the right time, with controlled complexity and measurable value delivery.

## Questions :

### 1. What is the importance of using a user persona in building the user story mapping?

User personas anchor product decisions in real user contexts rather than abstract assumptions.

In Agile product development, personas serve as decision filters. They influence backlog prioritization, acceptance criteria definition, and MVP slicing. Without personas, teams often optimize for technical possibilities instead of user value.

For RegimeLens, defining both non-technical investors and technically skilled but finance-inexperienced users clarified that:

- Simplicity outweighs automation
- Visualization outweighs raw model output
- Explanation outweighs prediction

The persona transformed story mapping from a feature listing exercise into a user-centered design process. It ensured that every user story answered the fundamental Agile question: “Who is this for, and what problem does it solve?”

Thus, personas operationalize empathy in Agile planning.

## **2. How do you think user story mapping helps you to achieve the triad of feasibility, engineering, and usability?**

User story mapping acts as a bridge between product strategy and technical execution.

### Feasibility

By organizing stories from high-level activities down to granular tasks, story mapping exposes dependencies, complexity layers, and risk concentrations. This allows early validation of whether the scope is technically achievable within constraints.

### Engineering Clarity

Story mapping decomposes large features into implementable slices. It creates natural sprint boundaries and supports incremental architecture design. Engineers can clearly see which components must exist before others.

### Usability

Because the map is structured around the user journey (left to right flow), it prevents teams from optimizing backend systems while neglecting user interaction flow. The product remains experience-driven rather than technology-driven.

In RegimeLens, story mapping ensured:

- Data retrieval precedes clustering
- Clustering precedes visualization

- Visualization precedes explanation

This preserved logical progression while balancing technical feasibility with user clarity.

User story mapping, therefore, operationalizes the Agile balance between business value, technical constraints, and human-centered design.

### 3. What is the significance of minimum viable product release?

The MVP is a strategic risk-management mechanism, not merely a reduced feature set.

Its significance lies in:

- Early validation of value proposition
- Reduction of technical and market risk
- Controlled investment of time and effort
- Enabling iterative learning cycles

For RegimeLens, the MVP focuses strictly on:

- Asset selection
- Basic indicator generation
- Regime clustering
- Color-coded visualization

This ensures the core hypothesis is validated:

“Users want clear and interpretable regime detection.”

Advanced features such as reinforcement learning, multi-asset comparison, or interactive dashboards are deferred because they increase complexity without validating the primary user need.

In Agile philosophy, the MVP transforms uncertainty into validated knowledge. It is the smallest experiment that tests product viability while preserving optionality for future iterations.

Through building the Product Vision and user story mapping for RegimeLens, I realized that clarity of user needs is more important than technical complexity. At first, I focused on advanced AI models, but defining the target personas helped me understand that interpretability and visualization are the real priorities.

Creating the backbone features structured the product logically and kept the focus on regime explanation rather than price prediction. Slicing the MVP was especially important because it forced me to identify what truly delivers value and remove unnecessary complexity.

This process strengthened my understanding of Agile thinking: start simple, focus on user value, and iterate gradually.

## Problem Statement

Financial markets are complex, data-intensive, and highly dynamic systems. Traditional financial analysis tools often assume that market behavior is stable over time. However, markets operate under different regimes (e.g., high volatility, low volatility, trending, or sideways periods). Models that ignore regime shifts often fail in real-world conditions due to non-stationarity and structural changes.

## Product Vision Board

| VISION                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Our vision is to build an explainable AI-based tool that allows users to observe financial markets through the lens of regime analysis. RegimeLens helps users identify and understand market regimes at the individual asset level rather than predicting price direction. |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                         |
| TARGET GROUP                                                                                                                                                                                                                                                                | NEEDS                                                                                                                                                                                                                                                                                | PRODUCT                                                                                                                                                                                                                                                                                                         | BUSINESS GOALS                                                                                                                                                                                                                                                                                                                          |
| 1, Non-technical individual investors<br>People who want simple explanations and visual insights.<br><br>2, Technical users without financial experience<br>People who understand AI / data but want to learn how it applies to finance.                                    | Many existing tools focus solely on prediction or trading signals. Users need a system that explains market behavior rather than automating decisions. They require: <ul style="list-style-type: none"><li>• Clear identification of market regimes</li><li>• Detection of</li></ul> | RegimeLens is an AI-based market regime analysis tool that: <ul style="list-style-type: none"><li>• Allows users to select any single asset (stock, index, or cryptocurrency)</li><li>• Automatically generates technical features</li><li>• Applies clustering techniques to identify market regimes</li></ul> | From a course and project perspective, RegimeLens enables our team to: <ul style="list-style-type: none"><li>• Demonstrate Agile development in a data-driven project</li><li>• Apply machine learning in a realistic and scoped manner</li><li>• Align user needs with technical implementation</li><li>• Build a foundation</li></ul> |

|  |                                                                                                                                                         |                                                                                                                                                                                                                                                       |                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|  | <p>regime transitions</p> <ul style="list-style-type: none"> <li>• Visual and interpretable explanations</li> <li>• Educational transparency</li> </ul> | <ul style="list-style-type: none"> <li>• Visualizes regimes directly on price charts</li> <li>• Provides human-readable summaries for each regime</li> </ul> <p>The system focuses on interpretability and education rather than trade execution.</p> | <p>extendable to forecasting, portfolio analytics, or risk management</p> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|

## Personas



### Sarah Ahmed

- 32 years
- Software Developer
- BSc Computer Science
- Toronto, Canada
- Beginner Investor

#### Bio

Sarah is a software developer who recently started investing in stocks and crypto.

She understands data and machine learning concepts but struggles to interpret financial market behavior.

She wants to make smarter decisions but does not trust black-box trading algorithms.

She prefers tools that explain "why" something is happening instead of simply predicting price direction.

#### Interests

- Data science
- AI explainability
- Financial independence
- Learning new analytical tools

#### Technical Comfort Level

- High technical ability
- Medium financial knowledge
- Wants educational transparency

#### Scenario:

Sarah opens RegimeLens to analyze Tesla stock.

She wants to understand:

- Whether the market is currently volatile or stable
- If the asset is trending or sideways
- When regime changes occurred historically

So that she can improve her understanding of market behavior before making any investment decision.

#### Pain Points

- Trading platforms only show indicators without explanation
- AI tools act as black boxes
- Difficult to detect regime shifts manually
- Too many confusing technical indicators
- Hard to know when volatility conditions change

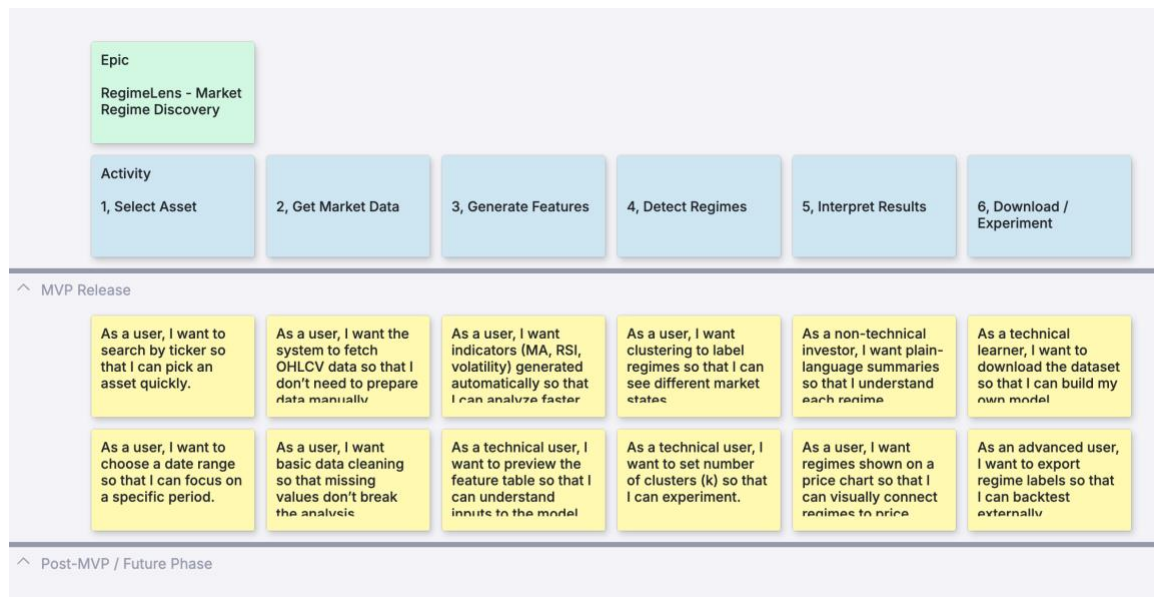
#### Goals/Needs

- A simple visual way to see market regimes
- Clear identification of regime transitions
- Human-readable explanations of each regime
- Understand why clustering labeled a regime
- Learn how AI applies to financial markets

Product: **RegimeLens** - Explainable AI Market Regime Tool

Persona: **Sarah Ahmed**

## User Story Mapping



## Product Backlog Items

| Id     | Name                        | User story                                                                                                   | How to test                                                                          | Importance | Estimate |
|--------|-----------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------|----------|
| PBI-01 | Asset Search                | As a user, I want to enter a ticker symbol so that I can analyze a specific asset.                           | Enter valid and invalid ticker; system loads valid asset and shows error for invalid | High       | 3        |
| PBI-02 | Symbol Validation           | As a user, I want the system to validate the ticker symbol so I know it exists.                              | Input fake ticker; system displays validation message                                | High       | 2        |
| PBI-03 | Historical Data Retrieval   | As a user, I want historical OHLCV data retrieved so that regime analysis can be performed.                  | Select date range; verify correct data is loaded                                     | High       | 5        |
| PBI-04 | Missing Data Handling       | As a user, I want missing values handled automatically so analysis remains accurate.                         | Introduce dataset with missing values; system fills or removes correctly             | Medium     | 3        |
| PBI-05 | Volatility Calculation      | As a user, I want volatility calculated so I can understand risk levels.                                     | Compare output with manual volatility calculation                                    | High       | 3        |
| PBI-06 | Moving Average Calculation  | As a user, I want moving averages generated to observe trend behavior.                                       | Verify MA formula correctness against sample data                                    | Medium     | 3        |
| PBI-07 | RSI Indicator               | As a user, I want RSI calculated so I can observe momentum conditions.                                       | Validate RSI against known dataset                                                   | Low        | 3        |
| PBI-08 | Regime Clustering           | As a user, I want the system to detect at least 3 market regimes automatically.                              | Run clustering; verify 3+ distinct labeled clusters appear                           | High       | 5        |
| PBI-09 | Regime Labeling             | As a user, I want regimes labeled consistently so I can interpret them easily.                               | Reload model; confirm same regime labels assigned logically                          | Medium     | 2        |
| PBI-10 | Regime Visualization        | As a user, I want regimes displayed on a color-coded price chart so I can visually understand market states. | Confirm chart shows distinct color segments by regime                                | High       | 5        |
| PBI-11 | Interactive Timeline        | As a user, I want to hover or zoom to inspect regime transitions over time.                                  | Use zoom/hover; verify data updates dynamically                                      | Medium     | 5        |
| PBI-12 | Regime Explanation Summary  | As a user, I want a plain-language explanation of each regime so I understand what it means.                 | Select regime; verify readable explanation appears                                   | High       | 3        |
| PBI-13 | Regime Transition Detection | As a user, I want to see when regime changes occur so I can understand structural shifts.                    | Check transition markers align with cluster changes                                  | High       | 3        |
| PBI-14 | Export Chart                | As a user, I want to export the chart so I can share analysis.                                               | Click export; file downloads successfully                                            | Low        | 2        |

## MVP – Minimum Viable Product

The MVP of RegimeLens is a web-based prototype that allows users to:

- Select a financial asset
- Automatically retrieve historical market data
- Generate basic technical indicators
- Apply a clustering model to identify market regimes
- Visualize detected regimes using a color-coded price chart

The MVP focuses on delivering clear and interpretable market state detection rather than advanced predictive trading strategies.

It prioritizes simplicity, transparency, and educational value for non-technical investors and technical learners without financial background.

## Research Summary

| Research Paper Link                                                         | Problem Statement Identified                                                                         | Tech Solutions Proposed                                          | Dataset Used in the Paper                     | Future Work                            |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------|----------------------------------------|
| DeepLOB: Deep Convolutional Neural Networks for Limit Order Books (Zhang et | Handcrafted features fail to capture complex spatial and temporal patterns in limit order book data. | CNN-based spatial feature extraction combined with LSTM temporal | High-frequency limit order book data from the | Evaluate robustness across markets and |



|                                                                                                                                                                                                         |                                                                                             |                                                                                             |                                                                    |                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------|
| al., 2018)<br>_FullLinks<br><a href="https://arxiv.org/abs/1808.03668">https://arxiv.org/abs/1808.03668</a>                                                                                             |                                                                                             | modeling in an end-to-end framework.                                                        | London Stock Exchange (LSE).                                       | incorporate realistic transaction costs.                                |
| FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading in Quantitative Finance (Yang et al., 2020)<br><a href="https://arxiv.org/abs/2011.09607">https://arxiv.org/abs/2011.09607</a> | Financial reinforcement learning lacks standardized benchmarks and reproducible evaluation. | Unified RL framework with standard environments, datasets, and evaluation metrics.          | US stock market OHLCV data (e.g., Dow Jones 30 constituents ).     | Improve generalization to unseen market regimes and extreme events.     |
| FinRL Contests: Benchmarking Data-Driven Financial Reinforcement Learning Agents (Yang et al., 2024)<br><a href="https://arxiv.org/html/2504.02281v3">https://arxiv.org/html/2504.02281v3</a>           | Inconsistent experimental settings make comparison of trading agents unreliable.            | Competition-style benchmark with fixed datasets, rolling windows, and standardized metrics. | Dow Jones 30 stock data with rolling train-validation-test splits. | Enhance robustness under non-stationary and volatile market conditions. |
| StockNet: Predicting Stock Price Movement with Twitter Sentiment (Xu & Cohen, 2018)<br><a href="https://arxiv.org/abs/1801.07890">https://arxiv.org/abs/1801.07890</a>                                  | Price-only prediction models ignore information embedded in social media sentiment.         | Neural models integrating historical prices with Twitter-based sentiment signals.           | StockNet dataset: historical prices aligned with Twitter data.     | Extend to multi-source text data and improve noise handling.            |
| BERTweet: A Pre-trained Language Model for English Tweets (Nguyen et al., 2020)<br><a href="https://arxiv.org/abs/2005.10200">https://arxiv.org/abs/2005.10200</a>                                      | Generic language models perform poorly on informal social media text.                       | Domain-specific transformer pre-training on large-scale Twitter data.                       | 850M English tweets collected between 2012–2019.                   | Adapt models for finance-specific sentiment analysis.                   |

|                                                                                                                                                                                                                                                                   |                                                                                     |                                                                              |                                                                  |                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|
| Stockformer: A Transformer-Based Framework for Stock Time Series Forecasting (2025)<br><a href="https://arxiv.org/abs/2502.09625">https://arxiv.org/abs/2502.09625</a>                                                                                            | Single-asset forecasting ignores cross-asset dependencies.                          | Multivariate Transformer architecture for financial time series modeling.    | Multi-asset stock price data from public financial APIs.         | Investigate dynamic retraining and improved temporal encoding.     |
| LSTM-Based Deep Learning Models for Nonlinear Financial Time Series Forecasting (Fischer & Krauss, 2018)<br><a href="https://www.sciencedirect.com/science/article/pii/S0378426618300235">https://www.sciencedirect.com/science/article/pii/S0378426618300235</a> | Linear models cannot capture nonlinear patterns in stock prices.                    | Long Short-Term Memory (LSTM) neural networks for sequential modeling.       | S&P 500 constituent stock price data.                            | Integrate macroeconomic and fundamental variables.                 |
| Temporal Fusion Transformers for Interpretable Multi-Horizon Time Series Forecasting (Lim et al., 2021)<br><a href="https://arxiv.org/abs/1912.09363">https://arxiv.org/abs/1912.09363</a>                                                                        | Standard sequence models struggle with long-term dependencies and interpretability. | Attention-based temporal fusion transformer architecture.                    | Multiple real-world financial and economic time series datasets. | Improve explainability and scalability for financial applications. |
| A Dual-Stage Attention-Based Recurrent Neural Network for Time Series Prediction (Qin et al., 2017)<br><a href="https://arxiv.org/abs/1704.02971">https://arxiv.org/abs/1704.02971</a>                                                                            | RNN models treat all historical inputs equally.                                     | Dual-stage attention mechanism to focus on relevant features and time steps. | Stock prices and technical indicator datasets.                   | Apply attention mechanisms to cross-market forecasting.            |
| Market Regime Detection Using Hidden Markov                                                                                                                                                                                                                       | Assuming a single market regime leads                                               | Hidden Markov Models for                                                     | Equity index return time series.                                 | Combine regime detection                                           |

|                                                                                                                                                                                                                                                                       |                                                   |                                                                     |                                                   |                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------|
| Models (Nystrup et al., 2015)<br><a href="https://www.sciencedirect.com/science/article/pii/S0377221715000833">https://www.sciencedirect.com/science/article/pii/S0377221715000833</a>                                                                                | to unstable trading strategies.                   | detecting latent market states.                                     |                                                   | with dynamic portfolio allocation.                            |
| Risk-Sensitive Reinforcement Learning Applied to Trading (Moody & Saffell, 2001)<br><a href="https://papers.nips.cc/paper/2001/file/4f6ffe13a5d75b2d6a39286b6e39d7c8-Paper.pdf">https://papers.nips.cc/paper/2001/file/4f6ffe13a5d75b2d6a39286b6e39d7c8-Paper.pdf</a> | Profit-only optimization ignores downside risk.   | Risk-adjusted reward functions in reinforcement learning.           | Simulated and historical financial market data.   | Design more realistic and practical risk measures.            |
| Time-Series Generative Adversarial Networks (TimeGAN) (Yoon et al., 2019)<br><a href="https://arxiv.org/abs/1907.05321">https://arxiv.org/abs/1907.05321</a>                                                                                                          | Limited historical data reduces model robustness. | GAN-based framework for generating realistic synthetic time series. | Financial and non-financial time series datasets. | Improve conditional generation and realism of synthetic data. |